Advanced Generative AI



Introduction to Generative AI Models



Engage and Think



- Imagine you're using a music streaming app like Spotify. One day, you notice a new feature where you can upload a short humming or a rough melody you recorded. The app then creates a full, professionally arranged song based on your input. It adds instruments, harmonizes, and even generates lyrics that fit the mood of your melody.

- How do you think this app can take a simple hum or rough melody and turn it into a complete, polished song with lyrics and accompaniment? What kind of technology might be behind this creative transformation?

Learning Objectives

By the end of this lesson, you will be able to:

- Apply autoencoders to compress and decode image data for improving website loading times in e-commerce
- Utilize GANs to create high-quality synthetic images for augmenting datasets in healthcare applications
- Implement transformers to improve the accuracy of language translation services in multilingual communication platforms
- Comprehend the principles of RAG and explore methods to enhance responses in customer support chatbots through the integration of retrieved and generated information



Generative AI Model Types

Types of Generative AI models

Generative AI encompasses various model types, each with distinct characteristics and applications.

- Generative adversarial networks (GANs)
- Recurrent neural networks (RNNs)
- Variational models
- Transformers
- Autoencoders

Types of Generative AI models

Autoencoders

Autoencoders, a type of neural network, compress and decode data for tasks such as image generation.

Recurrent neural networks

RNNs handle sequences and generate text with LSTM networks for context-based tasks.

Generative adversarial networks

GANs, with a generator and discriminator, produce realistic images like StyleGAN.

Types of Generative AI models

Transformers

Transformers efficiently handle sequences with self-attention, popular in natural language processing.

Variational models

Models represent data distribution, enabling sampling. Example: VAEs for diverse images.

Applications of Generative AI Model Types

Different types of Generative AI models are used in various fields.

Autoencoders

Used in image denoising, dimensionality reduction, and anomaly detection in healthcare

Variational models

Applied in image synthesis, data augmentation, and semi-supervised learning

Transformers

Essential for machine translation, chatbots, and text summarization

Recurrent neural networks

Applied in text generation, language translation, and speech recognition systems

Generative adversarial networks

Used for image-to-image translation and creating deep fake videos

Quick Check



Which type of generative AI model is primarily used for handling sequences and generating text?

A. Generative Adversarial Networks (GANs)

B. Recurrent Neural Networks (RNNs)

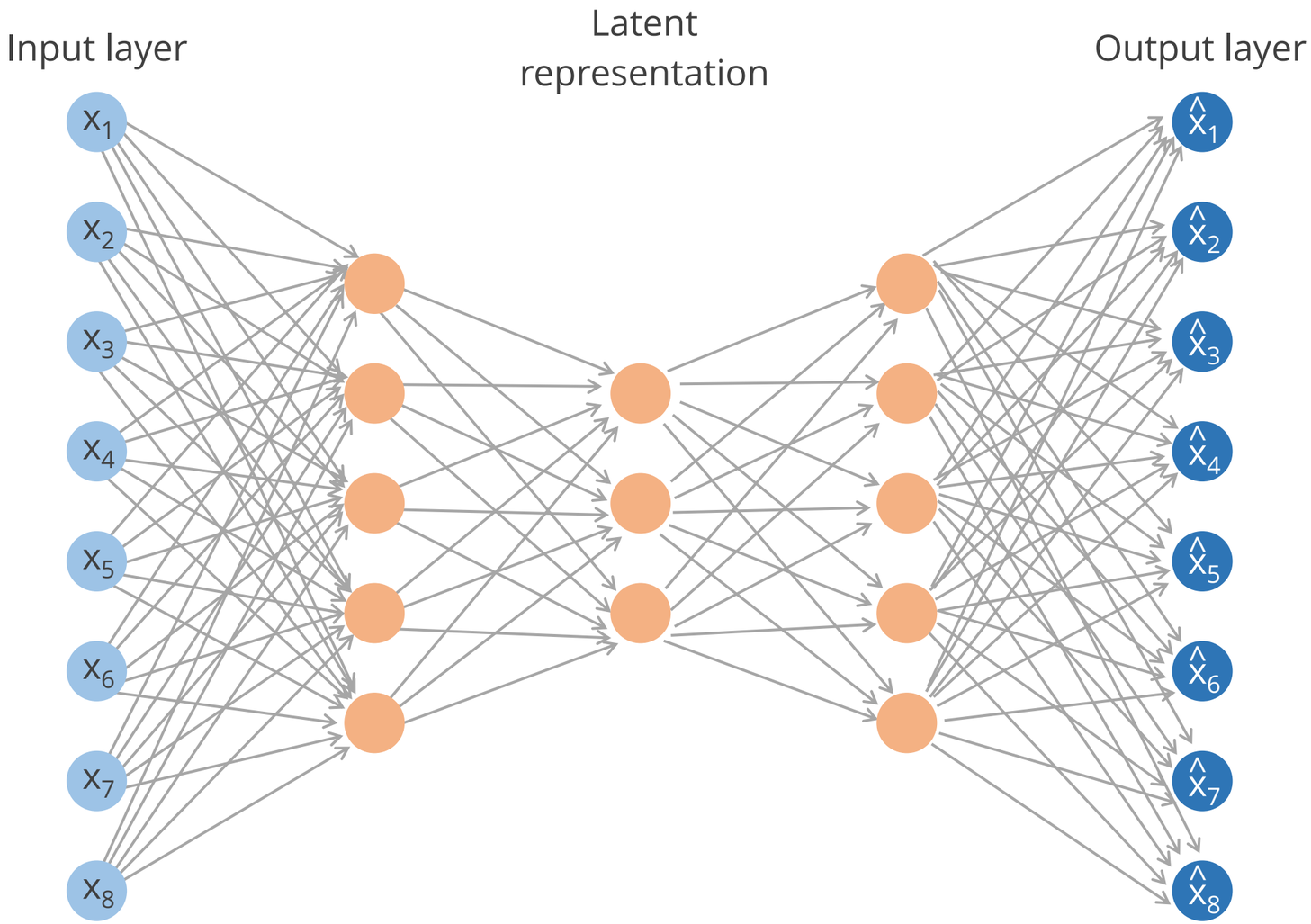
C. Autoencoders

D. Variational Models

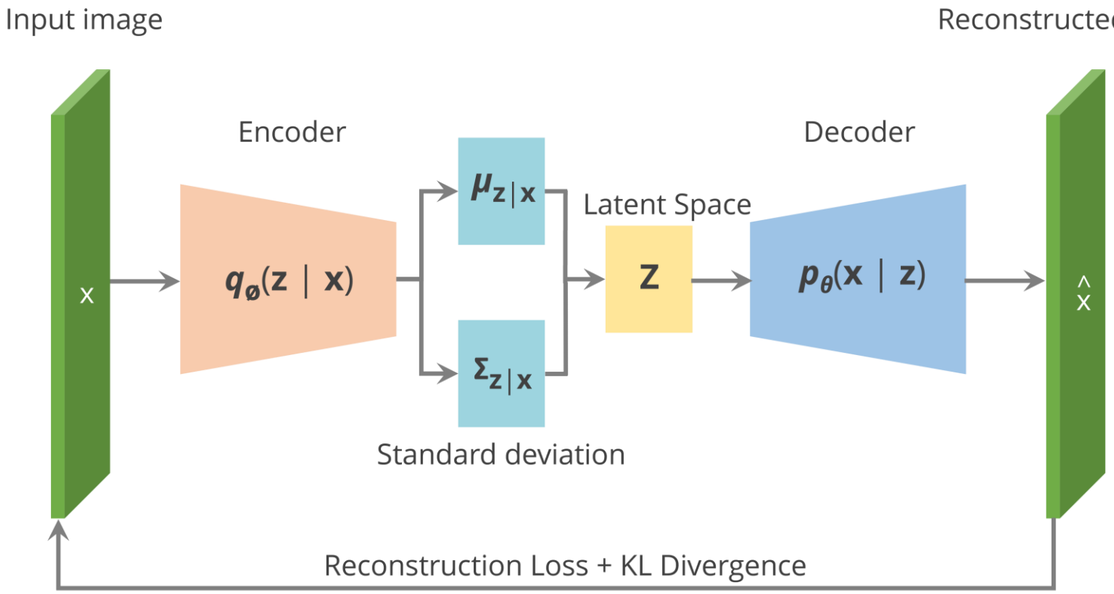
Introduction to Variational Autoencoders (VAEs)

Variational Autoencoders (VAEs)

Variational autoencoders (VAEs) are autoencoders with generative capabilities.

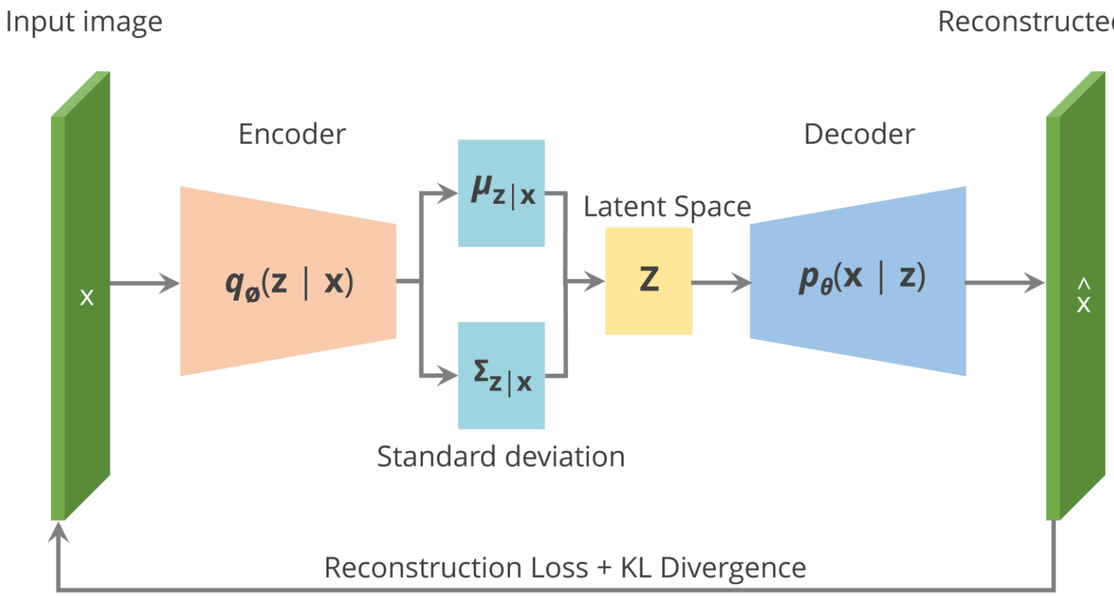


Variational Autoencoder Architecture



- Variational autoencoders (VAEs) are a type of generative model used in machine learning.
- They are proficient at data compression and generation tasks.
- VAEs consist of two primary components: an encoder and a decoder.
- The encoder compresses input data into a lower-dimensional representation called the latent space.

Variational Autoencoders Architecture



- The decoder then reconstructs data from this latent space.
- VAEs are unique because they introduce a probabilistic approach to data compression.
- This probabilistic element allows VAEs to generate new data samples from the latent space, offering a powerful tool for creative tasks and data generation.
- It incorporates an additional KL divergence term in its loss function, alongside the reconstruction loss, to enable probabilistic encoding.

Real-World Analogy: VAEs in Image Compression

VAEs work like a magic photo album:

- You take your photos (images) and put them into the album (encoding).
- The album captures the essence of your photos while reducing the size.
- When you want the original photo, the album helps recreate it (decoding).

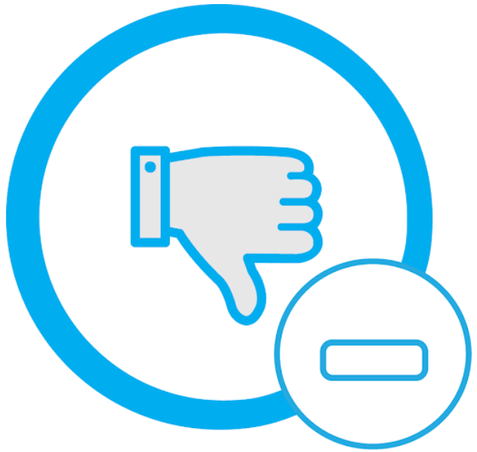
VAEs balance the need for efficient storage (smaller album) with the ability to recover the original images.

Note

In industries like e-commerce, using VAEs for image compression can save server space and improve website loading times without compromising image quality.

Drawbacks of VAE

The biggest drawback of VAEs is that they tend to produce blurry and unrealistic outputs.



Note

GANs are known for producing high-quality, sharp, and realistic outputs, particularly in image generation.

Quick Check



Question: In a Variational Autoencoder (VAE), what is the purpose of the latent space?

A. To store the original input data

B. To compress the input data for storage efficiency

C. To capture a probabilistic representation of the input data

D. To directly generate the output data from the input data

Introduction to Generative Adversarial Networks (GANs)

GANs: Introduction

Generative Adversarial Networks (GANs) are architectures in deep learning that use structures like convolutional neural networks for generative modeling.

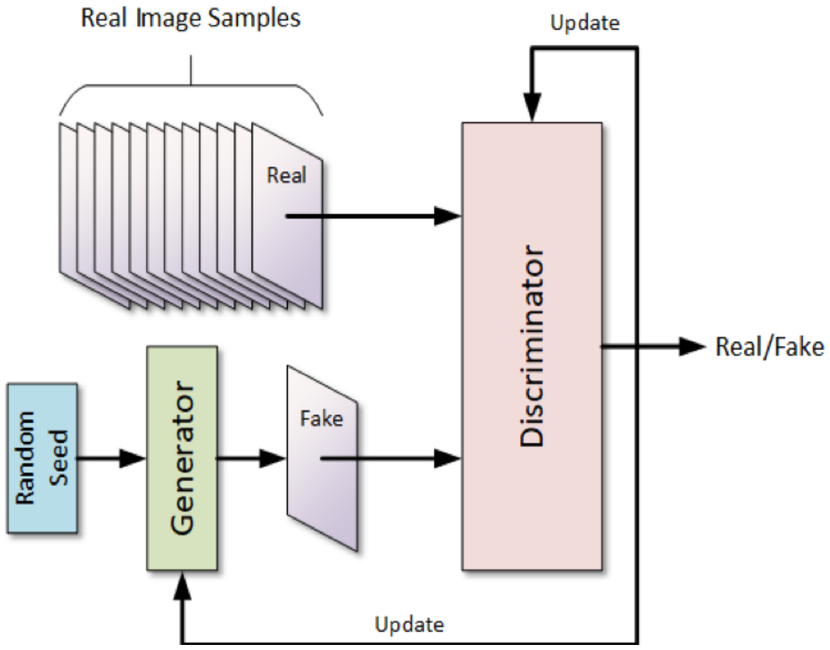


The images created above are imaginary.

- GANs generate highly realistic samples with sharp details and intricate features.
- They are effective in producing natural-looking images closely resembling input data.
- For example, a new image can be generated from an old training set.

Source: https://arxiv.org/abs/1906.00446?ref=assemblyai.com

Generative Adversarial Networks: Architecture

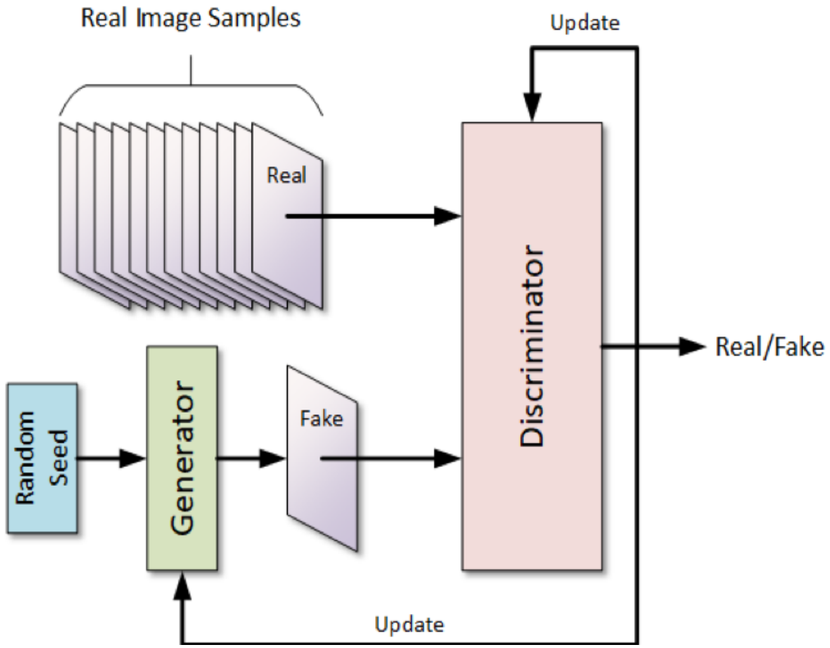


Generative adversarial networks (GANs) represent a revolutionary concept in Generative AI.

GANs consist of two neural networks: a generator and a discriminator.

The generator's role is to create synthetic data, while the discriminator's task is to differentiate between real and fake data.

Generative Adversarial Networks: Architecture



These networks engage in a zero-sum game, with the generator creating realistic data and the discriminator improving its fake detection, continuously enhancing both.

GANs have been remarkably successful in generating realistic images, text, and more, making them a cornerstone of Generative AI.

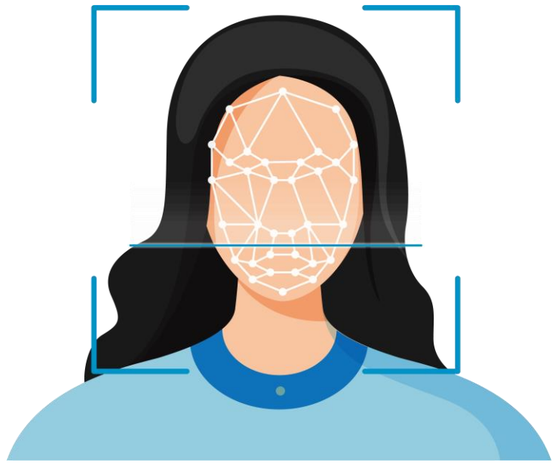
GANs have found applications in diverse fields, such as creating deepfake videos, enhancing image resolution, drug discovery, and material design.

Real-World Impact of GANs: Deepfake Technology

Let's delve into a notable real-world example: Deepfake technology

Deepfakes result from the remarkable capabilities of GANs.

They involve superimposing one person's face onto another person's body in videos, making it look convincingly real.



Real-World Impact of GANs: Deepfake Technology

GANs, with their adversarial training, excel at creating these forgeries.

While deepfake technology has raised concerns about misinformation and privacy, it also has promising applications in the entertainment and film industry.

Note

GANs continue to evolve and are making waves in fields like healthcare, art, and more, demonstrating their versatility and creative potential.

Quick Check



Question: In the architecture of a Generative Adversarial Network (GAN), what is the role of the discriminator ?

A. The discriminator generates new image data from a noise vector.

B. The discriminator guides the generator to produce more realistic images by providing feedback.

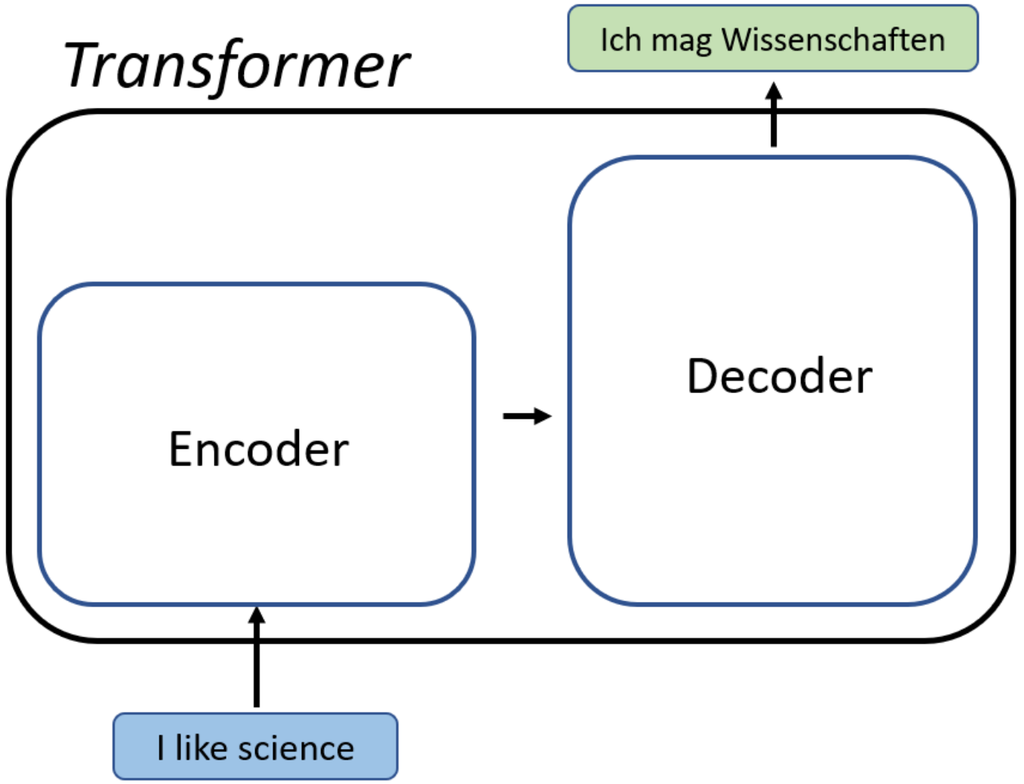
C. The discriminator classifies input images as real or fake.

D. The discriminator optimizes the noise vector to improve image generation.

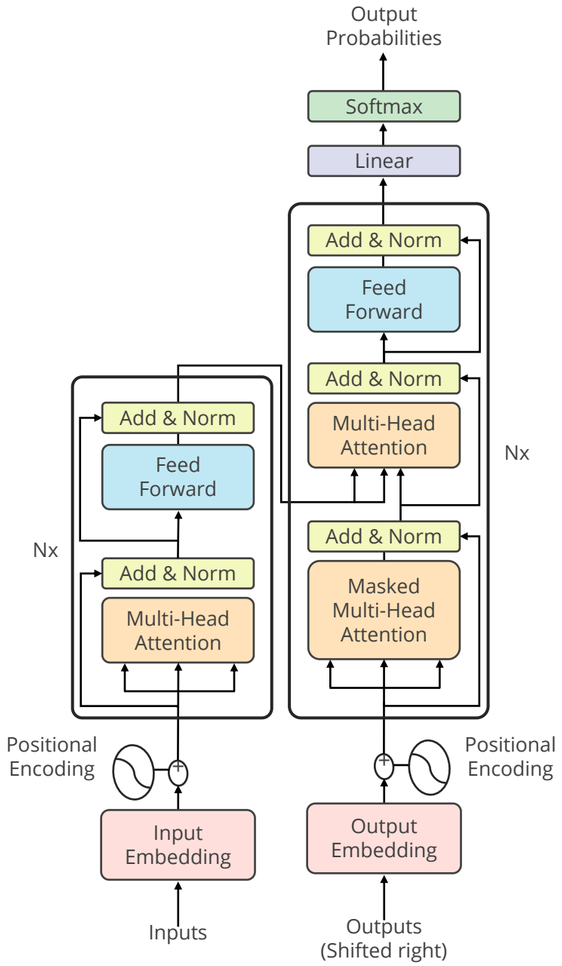
Introduction to Transformer-Based Models

Transformer-Based Models

Transformer-based models revolutionize language processing, enabling advanced tasks such as translation, summarization, and natural language understanding.



Transformer-Based Models

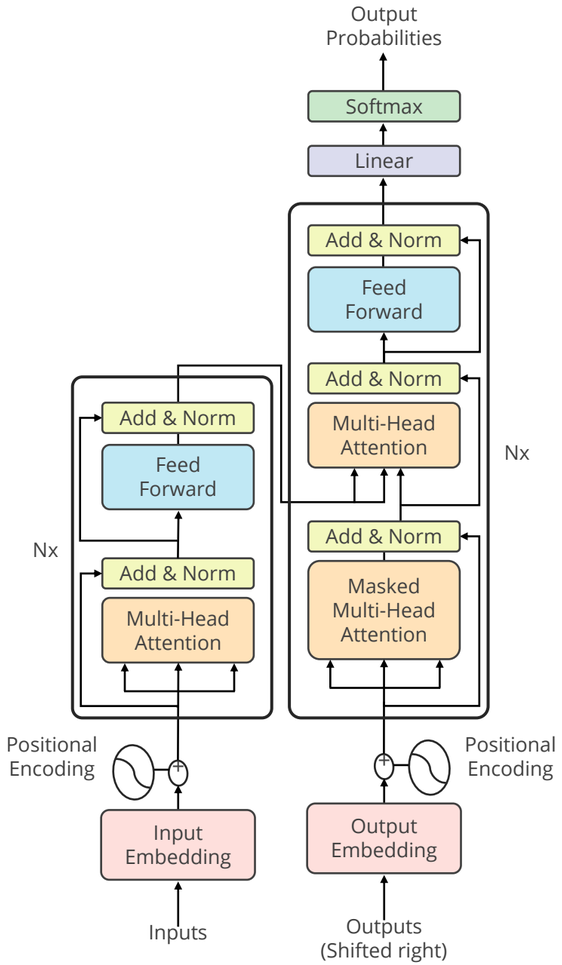


Transformer-based models are a revolutionary development in the field of natural language processing (NLP).

They were introduced in the paper ***Attention Is All You Need*** by Vaswani and others in 2017.

Unlike traditional sequence models, transformers do not rely on recurrent layers, making them highly parallelizable.

Transformer-Based Models



Transformers utilize a mechanism called self-attention to process input data in parallel, capturing dependencies between words in a sentence.

This self-attention mechanism allows transformers to outperform previous models in various NLP tasks.

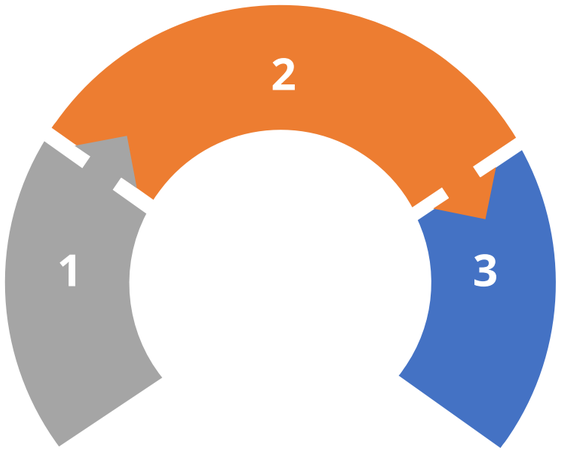
Transformer-based models, such as the GPT-3 and BERT, have significantly impacted the way one understands and generate human language.

Working of Attention Mechanism

Below are the working stages of an attention mechanism:

Assigning Weights
Models assign weights to input elements based on their relevance to a specific context or query.

Focusing on Relevant Information
Attention mechanisms enable models to selectively focus on relevant parts of the input data when generating output.



Contextual Computation
This weighted sum or context vector captures crucial information for the model's decision-making.

Self-Attention Mechanism

Types of Attention Mechanism

Various attention mechanisms are tailored for different applications, and three such types are the following:

Additive Attention

- Model creates a weighted sum of input elements by applying learned weights or parameters.
- This method calculates attention by learning the importance of each element in the input sequence.

Types of Attention Mechanism

Various attention mechanisms are tailored for different applications, and three such types are the following:

Multiplicative Attention

- Model generates attention weights through element-wise multiplication between the input elements and a learned parameter vector.
- This approach allows the model to capture complex interactions between elements.

Types of Attention Mechanism

Various attention mechanisms are tailored for different applications, and three such types are the following:

Self Attention

- Model attends to different positions of the input sequence by comparing each position with all other positions.
- It involves comparing each element with every other element in the sequence, including itself, to reweigh their importance based on contextual relevance.

Understanding Self-Attention

Self-attention, a key component in Natural Language Processing (NLP), enables the network to focus on particular words or phrases in a sentence for improving context understanding.

While reading a novel, you simultaneously focus on the current page and recall earlier events, characters, and clues.

Example:

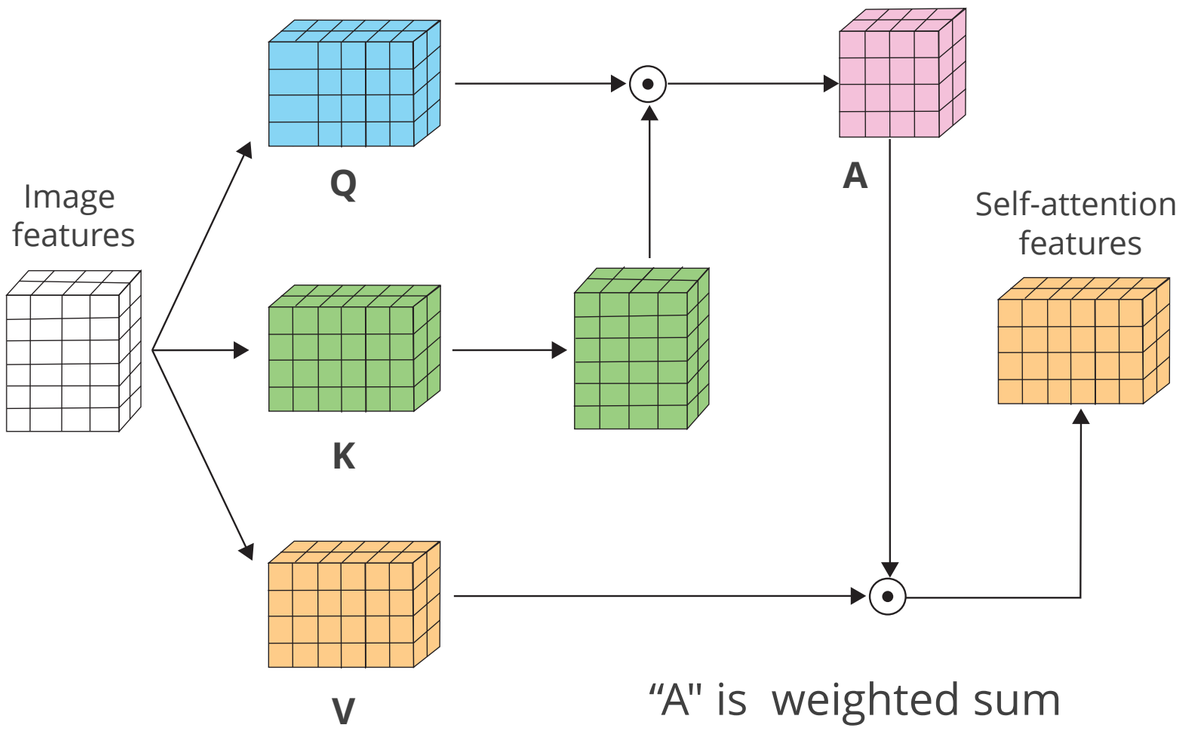


You are reading a mystery novel.

This context helps you understand the story better and predict future events.

Self-Attention

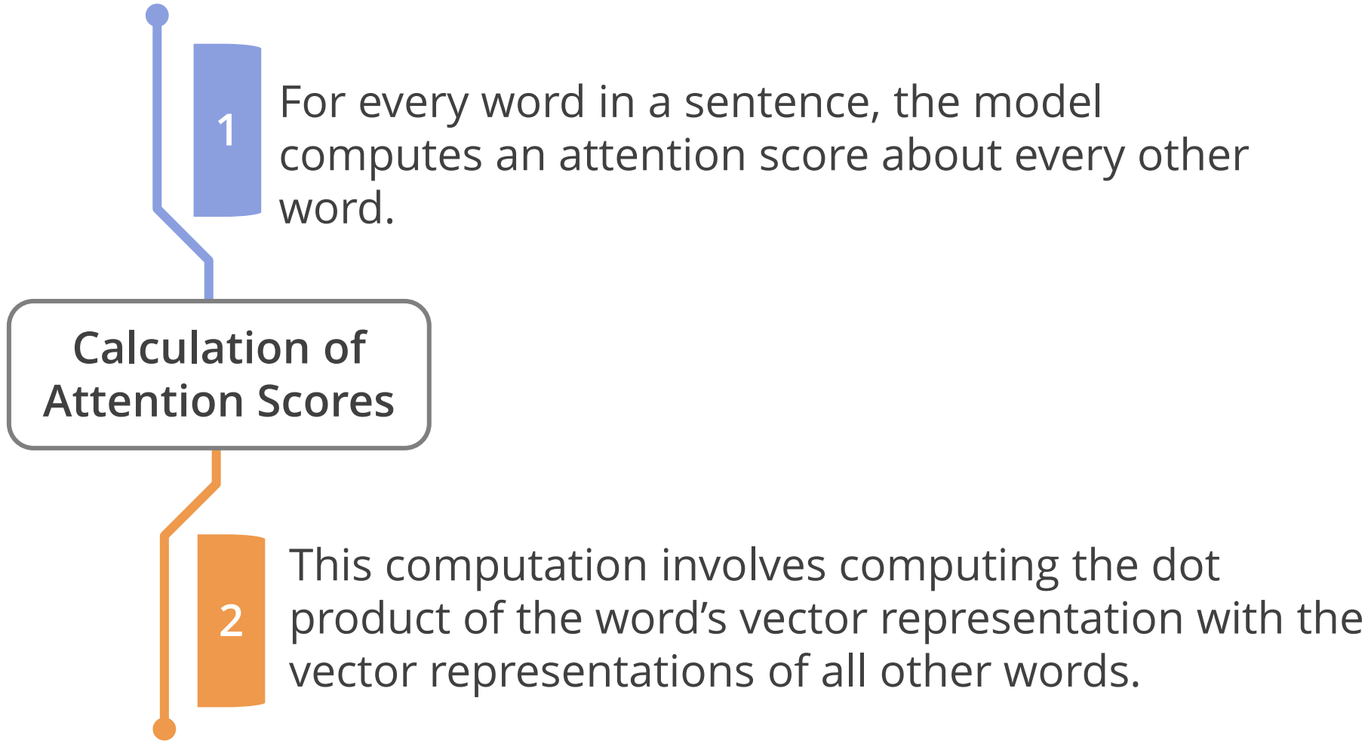
In the self-attention mechanism, the input image feature is defined as Query (Q), Key (K), and Value (V).



The attention mechanism is to calculate the attention weight between the query and key and then used to strengthen the value.

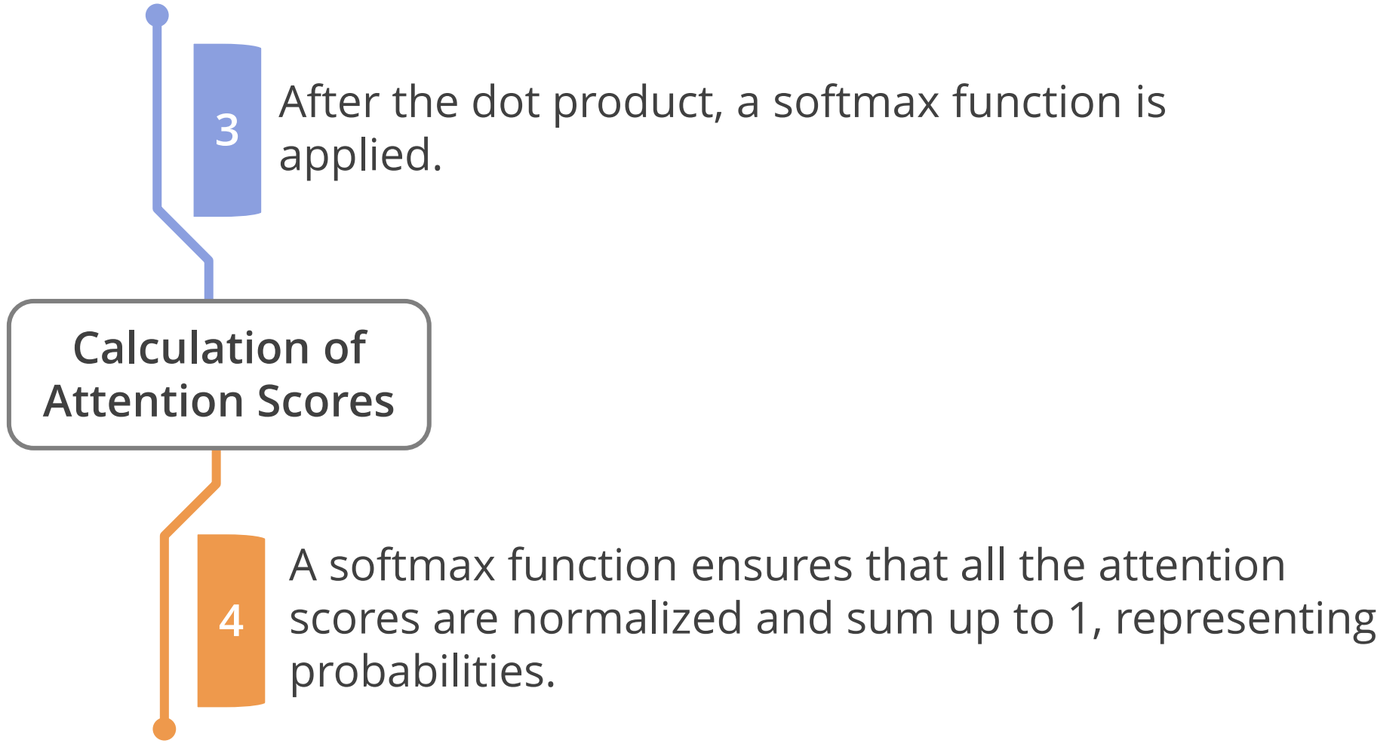
Mechanics Behind Self-Attention

In the self-attention mechanism, attention scores are crucial, determining the importance assigned to each word in a sentence during the encoding process.



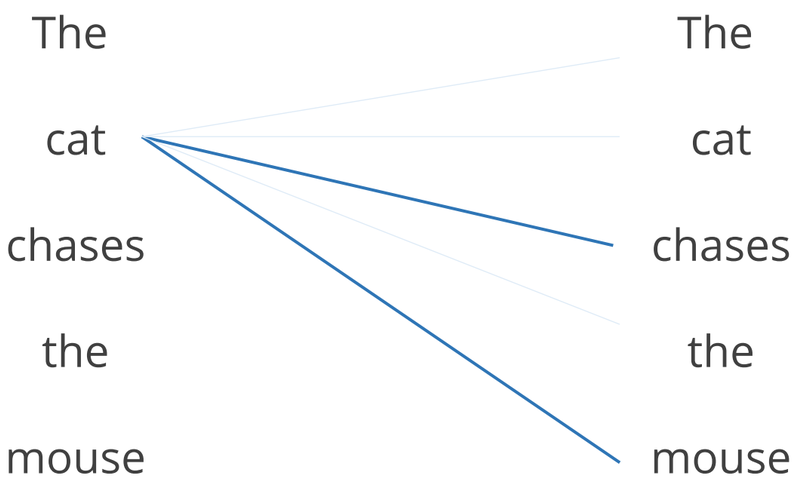
Mechanics Behind Self-Attention

In the self-attention mechanism, attention scores are crucial, determining the importance assigned to each word in a sentence during the encoding process.



Mechanics Behind Self-Attention

The attention scores are based on the relative positions and semantic similarities of words, with those closer in position or more similar semantically receiving higher scores.



Example: The cat chases the mouse.

- The word "cat" has semantic connections to both "chases" and "mouse."
- As a result, the pairs "cat-chases" and "cat-mouse" achieve high attention scores.
- This is because "cat" performs the action "chases," and "mouse" is the action's object.
- The model identifies these relationships and assigns higher attention scores accordingly.

Quick Check



Question: What is the primary purpose of the self-attention mechanism in the context of NLP models?

A. To reduce the computational complexity of the model.

B. To assign higher attention scores to words that are semantically similar or closer in position.

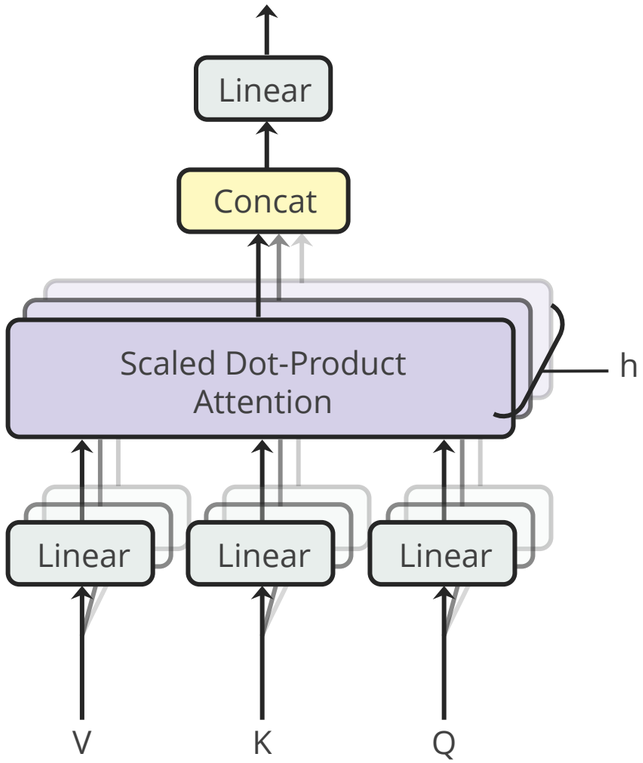
C. To increase the size of the model's vocabulary.

D. To translate sentences from one language to another.

Multi-Head Attention Mechanism

Multi-Head Attention

It is an extension of the self-attention mechanism that enables a model to focus on various parts of the input sequence at different times.



- Multi-head attention employs independent linear layers for keys (K), queries (Q), and values (V).
- It combines each attention head's output and processes it through a non-linear function.
- It enables the model to focus on different data positions.
- It treats inputs as sets of elements rather than as sequences.

h = Parallel attention layers or heads

Mechanics Behind Multi-Head Attention

Let’s understand the steps by considering a simple sentence: "The cat sat on the mat.”

Splitting the input

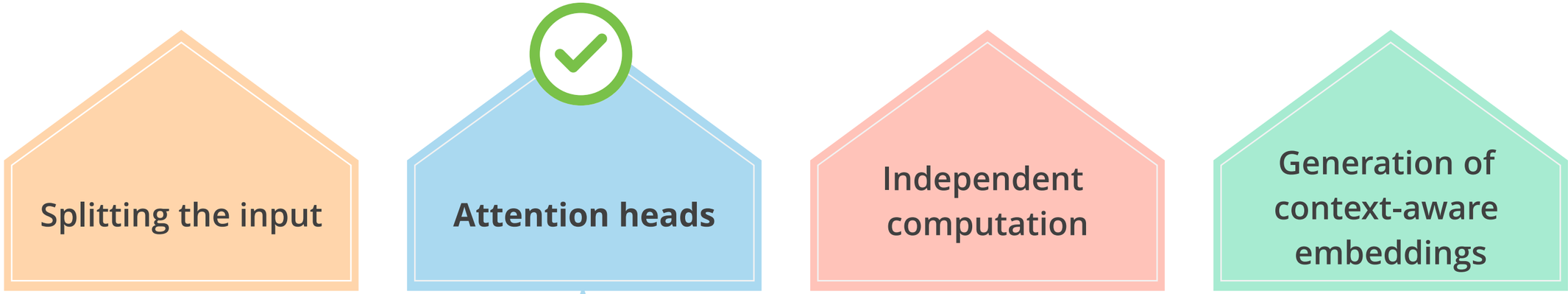
Attention heads

Independent computation

Generation of context-aware embeddings

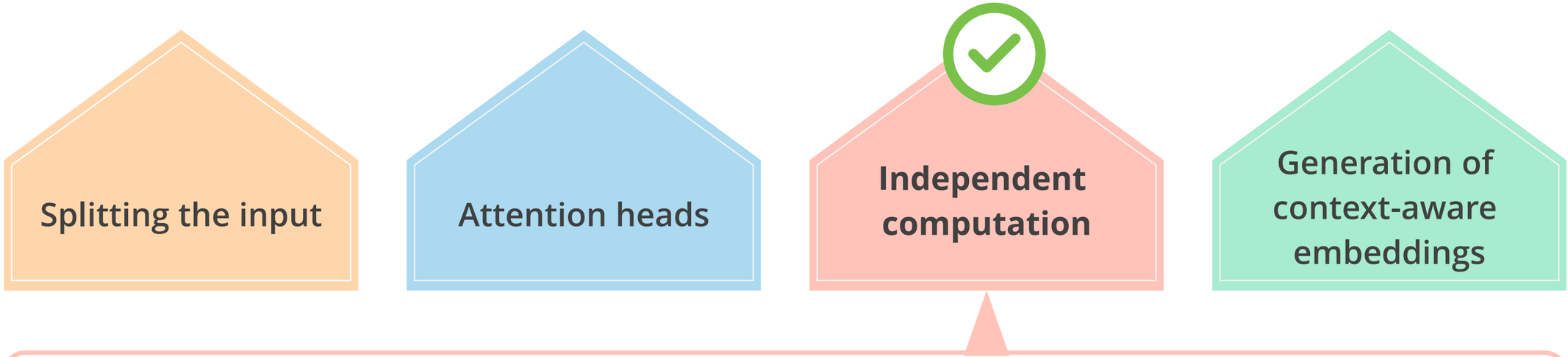
- Assume you use a two-head attention mechanism.
- First, transform the input embeddings of each word in the sentence using two distinct linear transformations.
- This process yields two sets of transformed inputs.
- Each set provides a unique perspective on the sentence.

Mechanics Behind Multi-Head Attention



- Each transformed input undergoes processing by a separate self-attention mechanism known as an **attention head.**
- The first attention head focuses on syntactic features, such as word order.
- The second attention head concentrates on semantic features, such as word meaning.

Mechanics Behind Multi-Head Attention



- Every attention head operates independently.
- Each attention head calculates attention scores by comparing each word in its input with every other word.
- For instance, when processing **cat**:
    a. The first attention head might assign high scores to **sat** and **mat**, due to their syntactic relation to a **cat**.
    b. The second attention head could give a high score to **the**, commonly found before nouns.

Mechanics Behind Multi-Head Attention

Splitting the input

Attention heads

Independent computation

Generation of context-aware embeddings

- Attention heads generate new embeddings for words using attention scores.
- For a word like **cat**, its new embedding is a weighted sum of all sentence word embeddings.
- These weights come from the attention scores.
- This method ensures that each word's embedding captures its meaning and sentence context.

Real-World Impact of Transformers: Language Translation

One practical example of transformer-based models is their use in language translation.

Traditional translation models required a fixed vocabulary and struggled with context in long sentences.

Transformers, however, can handle translations at the word level and maintain context effectively.



Real-World Impact of Transformers: Language Translation

They have powered services like Google Translate, improving the quality and fluency of translations.

This has made cross-language communication more accessible and accurate, benefitting international business, travel, and online content.

Note

Transformer-based models continue to revolutionize NLP and have a wide range of applications, including chatbots, content summarization, and sentiment analysis.

Quick Check



Which component of Transformer models enhances their ability to understand the context by allowing various perspectives on the data?

A. Multi-head attention

B. Dividing attention heads

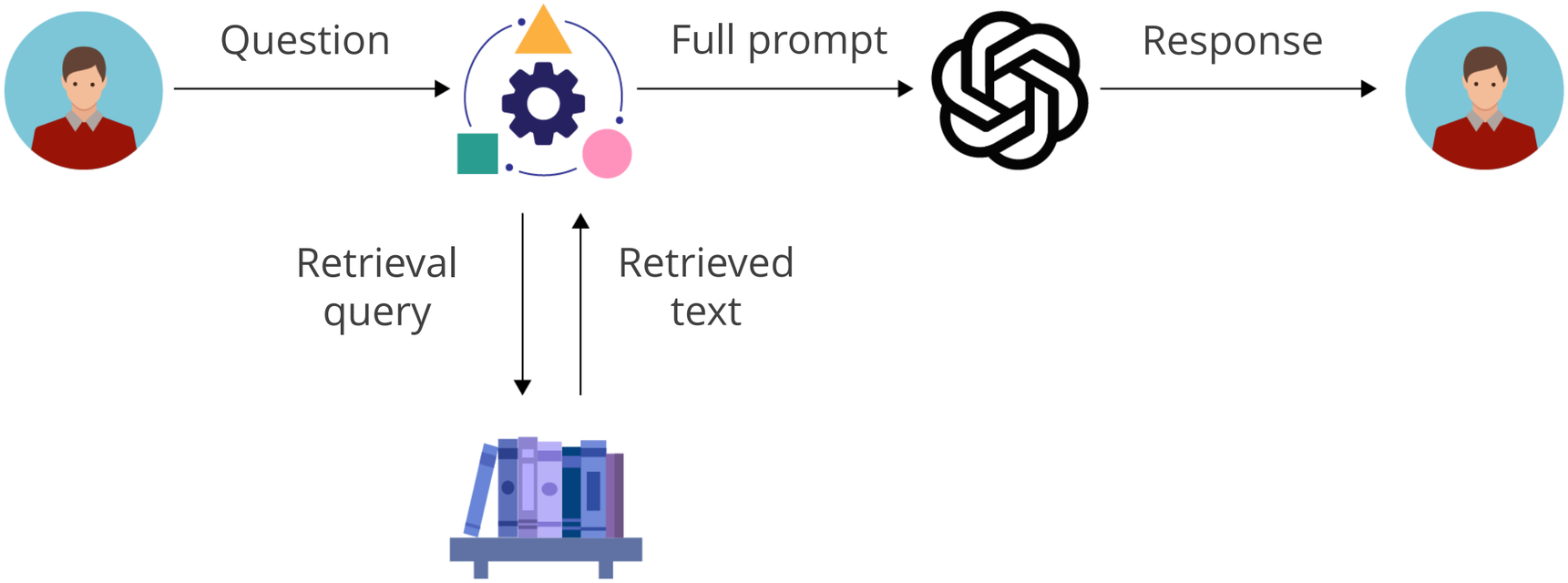
C. Attention mechanism

D. Transformer architecture

Retrieval Augmentation Generation (RAG)

Retrieval Augmentation Generation (RAG)

RAG combines retrieval and generation methods for enhanced natural language processing, leveraging preexisting knowledge while generating contextually relevant information.



Retrieval Augmentation Generation (RAG)

Retrieval augmentation generation (RAG) is an advanced concept in natural language processing (NLP).

RAG combines elements of retrieval-based and generative models to enhance AI's capabilities.

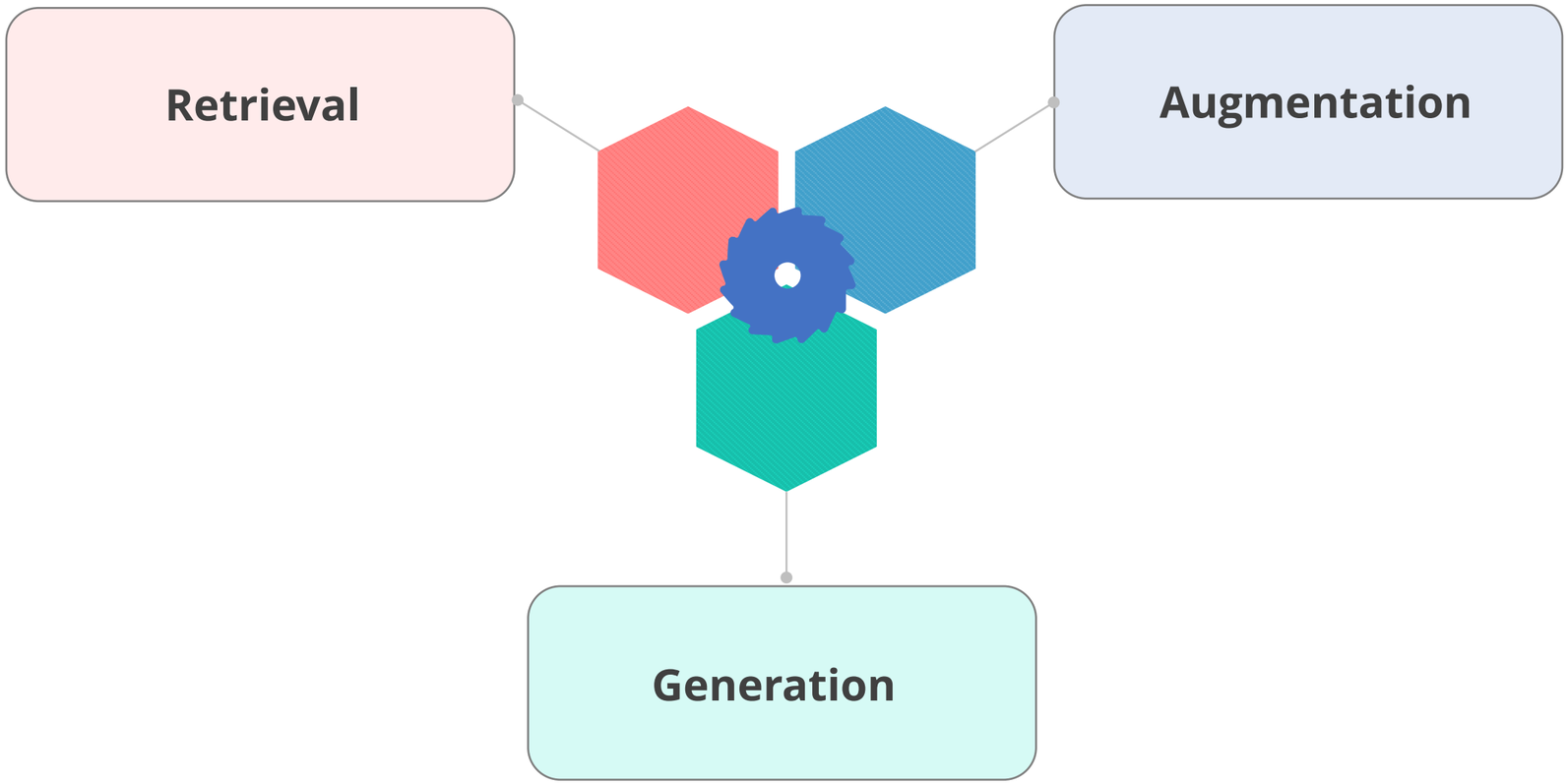
RAG is a framework that leverages both retrieval and generation components.

It combines the strengths of models like BERT for retrieval and GPT for generation.

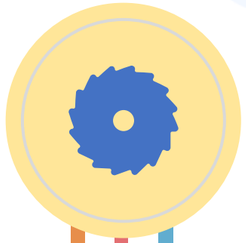
RAG aims to address limitations of purely generative or retrieval-based models.

Components of RAG

RAG consists of three main components:



Components of RAG



Retrieval Utilizes pre-trained models to fetch relevant information from large datasets

Augmentation Enhances retrieved information by paraphrasing, summarizing, or simplifying

Generation Produces responses or content based on the augmented information

Why RAG Matters?

RAG overcomes limitations of purely generative or retrieval-based systems.

RAG excels in tasks such as answering questions, chatbots, and generating content.

RAG (Retrieval-Augmented Generation) improves response quality and information accuracy by combining retrieval mechanisms with generative models.

Quick Check



Which component of Retrieval augmented generation (RAG) is responsible for enhancing retrieved information by paraphrasing, summarizing, or simplifying?

A. Retrieval

B. Augmentation

C. Generation

D. None of the above

Choice of Retriever

Choice of Retriever

Retrieval Augmented Generation (RAG) empowers Large Language Models (LLMs) to access external knowledge sources, extending their capabilities beyond their pre-trained weights.

RAG utilizes a retriever, which acts as a knowledge scout. This retriever searches external knowledge bases, such as databases, to find relevant documents or information.

Choice of Retriever

The retrieval component of RAG searches the database to find relevant documents or pieces of information that can provide context and factual accuracy to the generated response.

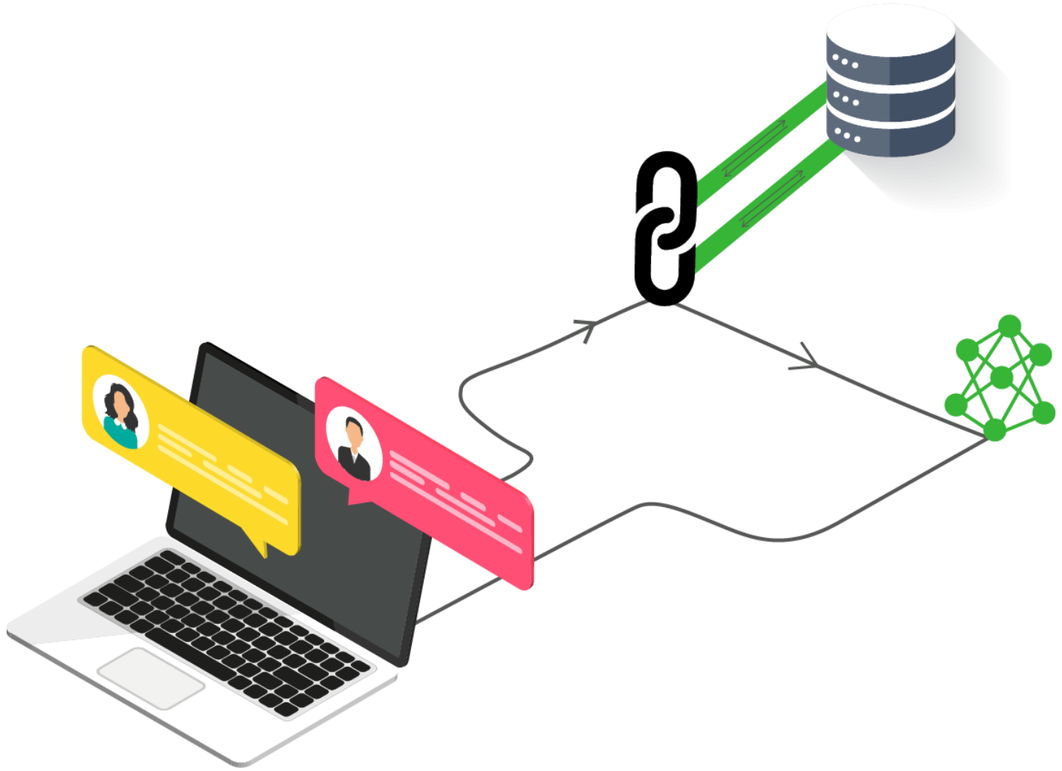
Vector database: For semantic retrieval, queries are embedded using models like BERT for dense embeddings or TF-IDF for sparse embeddings. The retriever searches with term frequency to find relevant information.

Graph database: This database structure is employed for knowledge retrieval when data relationships are critical.

Regular SQL database: For structured data, such databases offer a powerful way to access and augment the LLM's knowledge base.

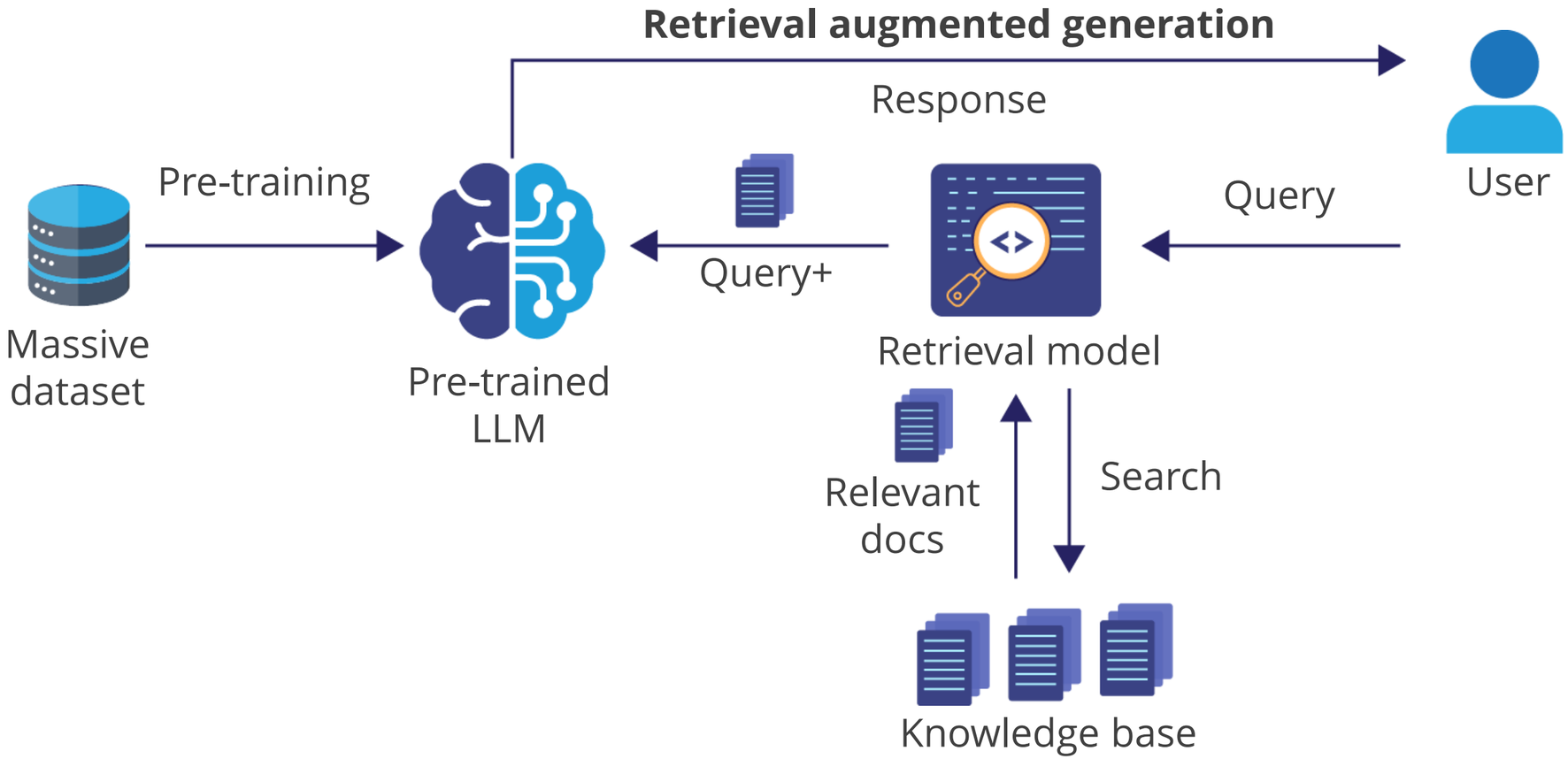
Choice of Retriever

RAG effectively marries the capabilities of an LLM with external knowledge sources, making it a potent tool for context-aware and knowledge-enriched text generation and understanding.



Process of Retrieval Augmented Generation (RAG)

Retrieval augmented generation (RAG) is a multi-step process that seamlessly integrates external knowledge with LLMs.



Process of Retrieval Augmented Generation (RAG)

RAG is an architectural approach that improves the efficiency of the LLM by following certain process which involves:

Vector database creation:

RAG begins by converting data into vectors, which is essential for quick retrieval.

User input

Users ask questions in plain language, often seeking answers or completion.

Information retrieval:

Database is scanned for segments like user’s query for matching.

Combining data:

Chosen database segments combine with user's query for an enriched prompt.

Generating text:

Expanded prompt with added context guides LLM to create relevant response.

Process of Retrieval Augmented Generation (RAG)

This process seamlessly integrates external knowledge sources, enhancing the large language model's ability to provide more accurate and contextually rich responses to user queries.



Real-World Applications of RAG

RAG has practical applications in various fields:



Customer support chatbots that provide accurate and context-aware responses



Search engines that understand user queries and generate informative snippets



Content generation for news articles, product descriptions, and more

Quick Check



Which type of database structure is typically employed for knowledge retrieval when data relationships are crucial?

A. Vector database

B. Graph database

C. Regular SQL database

D. NoSQL database

Guided Practice



Overview **Duration: 20 minutes**

In this activity, test your understanding of the different generative model types covered in this module. A scenario will be given with a list of generative model types, and you will have to choose the most suitable one for the scenario. Also, explain why you chose that model type and how it works.

Note

Please download the solution document from the Reference Material Section and follow the document for step-by-step execution.

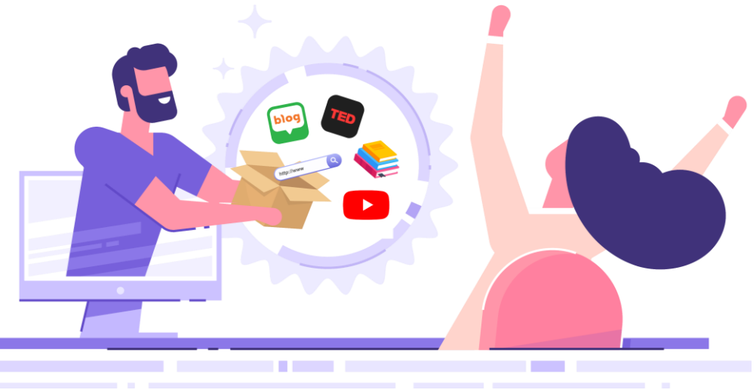
GUIDED PRACTICE

Key Takeaways

- Generative AI includes different types of models, each with unique traits and uses.
- Autoencoders, a neural network type, condense and decode data, serving tasks like generating images.
- Generative AI revolutionizes technology interaction but demands ethical attention to aspects like privacy and misuse.
- RAG enhances language processing by blending retrieval and generation, using existing knowledge for contextually relevant information.



Additional Resources



- Nah, F. F., Zheng, R., Cai, J., Siau, K., & Chen, L. (2023). Generative AI and ChatGPT: Applications, challenges, and AI-human collaboration. Journal of Information Technology Case and Application Research, 25(3), 277–304. https://doi.org/10.1080/15228053.2023.2233814

- Bandi, A., Adapa, P. V. S. R., & Kuchi, Y. E. V. P. K. (2023). The power of Generative AI: a review of requirements, models, Input–Output formats, evaluation metrics, and challenges. Future Internet, 15(8), 260. https://doi.org/10.3390/fi15080260

Q&A

